

- 24** A charged particle is projected with velocity v into an evacuated region. A uniform electric field of field strength E and a uniform magnetic field of flux density B are applied perpendicular to each other in the same evacuated region.

If the velocity v is to stay constant, what is its magnitude and direction?

	magnitude	direction
A	$\frac{B}{E}$	parallel to B
B	$\frac{E}{B}$	parallel to E
C	$\frac{B}{E}$	perpendicular to both B and E
D	$\frac{E}{B}$	perpendicular to both B and E

- 25** A current carrying wire is placed below a vertical coil in its plane with current flowing